# MediAssist

AI-Powered Prescription Scanner & Medicine Reminder System

**MCA Minor Project Proposal**

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## Objective

To design and develop a web-based medical assistance application that scans prescriptions, extracts medicine names and dosage frequency, stores them securely, sends real-time reminders for each dose, and helps users find the medicines online and at nearby medical stores using their detected location.

## Synopsis

Patients, especially the elderly, often struggle to read prescriptions, interpret dosage notations such as OD, BD, TDS or 1-0-1, and remember when to take each medicine. Missed or incorrect doses reduce treatment effectiveness, and finding the right medicines across stores wastes time. This project builds MediAssist, an application where a user uploads or captures a photo of a prescription. The system preprocesses the image, applies OCR to extract text, and parses it to identify each medicine, its strength, frequency and duration, converting notations into a clear daily schedule. The user confirms or corrects the extracted details before they are saved to a personal account. A scheduler then triggers real-time browser notifications at the correct dose times, with options to mark a dose as taken, snooze or skip it. For every medicine, the app suggests online pharmacy search links and uses the browser's geolocation to show nearby medical stores on an interactive map. MediAssist is an assistive tool and does not replace a doctor's or pharmacist's advice.

## Tech Stack

1. **Frontend / UI** - HTML, CSS, JavaScript (Service Workers for notifications)
2. **Backend** - Python (Flask / FastAPI)
3. **OCR & Image Processing** - Tesseract / EasyOCR, OpenCV
4. **Prescription Parsing** - spaCy, Regular Expressions, RapidFuzz (medicine name matching)
5. **Reminders** - APScheduler, Web Push / Notifications API
6. **Location & Maps** - HTML5 Geolocation API, OpenStreetMap (Overpass) with Leaflet
7. **Storage** - SQLite / PostgreSQL

## Expected Outcome

A working end-to-end web application where a user uploads a prescription and receives a clear medicine schedule, automatic dose reminders, and online and nearby store recommendations for their medicines - reducing missed doses and the time spent understanding prescriptions and searching for medicines.